Student ID (SID):
Student's Name:

CSCE-4613/5613 Artificial Intelligence

Spring 2025

Final Exam

Form A

Date: Tuesday, May 6, 2025

Duration: 12:45 PM - 2:45 PM

Instructions

1. It is an individual exam.
2. You have 120 minutes. You may not leave during the last 15 minutes of the exam.
3. Do NOT open exams until told to. Write your SIDs in the top left corner of every page.
4. If you need to go to the bathroom, bring us your exam, phone, and SID. We will record the time.
5. In the interest of fairness, we want everyone to access the same information. To that end, we will not be answering questions about the content.
6. The exam is closed book, closed laptop, and closed notes, except your TWO-page double-sided cheat sheet. You are allowed a non-programmable calculator for this exam. Turn off and put away all other electronics.
7. Mark your answers - ON THE EXAM IN THE ANSWER AREAS. We will not grade anything on scratch paper.

Student ID (SID):
Student's Name:

Question 1: (1 point)

Given a binary 2D matrix, find the number of islands. A group of connected 1s forms an island. (Two cells are called connected in an island if there is a path between them. Two cells sharing at least one edge will connect to each other.)

Input: mat[][] = {{1, 1, 0, 1, 0},
{0, 1, 0, 0, 1},
{1, 1, 0, 1, 1},
{0, 0, 0, 1, 0},
{1, 1, 1, 1, 0}}

What algorithms can be used to find the number of islands? Please name at least two algorithms.

Question 2: (1 point)

How many islands are in of matrix in Question 1?

(A) 1 island
(B) 2 islands
(C) 3 islands
(D) 4 islands

Question 3: (1 point)

On a Tree, will Breadth First Search (BFS) and Depth First Search (DFS) produce the same results? A tree is a graph that does not contain any cycles.

(A) Yes
(B) No

Student ID (SID):
Student's Name:

Question 4: (1 point)

Consider performing BFS and DFS on a connected, undirected graph starting from the same node. Which of the following statements is **always true**?

(A) The order in which nodes are visited by BFS and DFS is the same.

(B) The BFS tree is a subgraph of the DFS tree.

(C) BFS produces a minimum-depth tree (shortest path tree), while DFS does not guarantee minimum depths.

(D) DFS always finds a path with the fewest edges between the starting node and any other node.

Question 5: (1 point) **(for graduate students only)**

When running Depth-First Search (DFS) on a **directed graph**, edges encountered during the traversal can be classified into four types:

- **Tree Edge:** An edge that discovers a new vertex (i.e., it leads to a vertex not yet visited).
- **Back Edge:** An edge that connects a vertex to one of its ancestors in the DFS tree (going backward along the search path).
- **Forward Edge:** An edge that connects a vertex to a descendant (already visited but not the direct child).
- **Cross Edge:** An edge that connects two vertices in different DFS trees or branches, not ancestor/descendant.

Given these definitions, which of the following statements is **correct**?

(A) In an undirected graph, DFS may produce back edges but never forward or cross edges.

(B) A forward edge during DFS traversal always indicates the existence of a cycle.

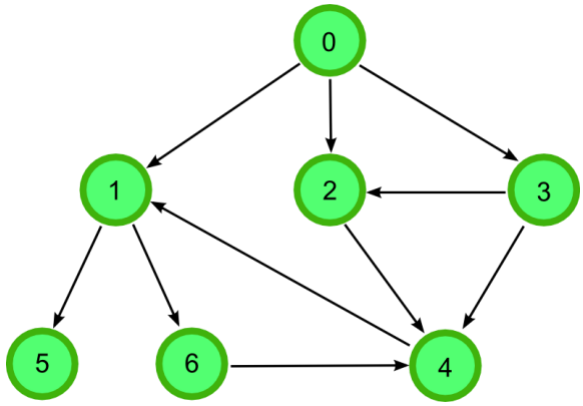
(C) A back edge detected by DFS indicates the presence of a cycle in a directed graph.

(D) If there are no cross edges during DFS traversal, the graph must be acyclic.

Student ID (SID):
Student's Name:

Question 6: (1 point)

Write the order of nodes visited by DFS on this graph (starting from 0). In addition, we assume that at a given step, if the vertex u can go to either v or v', it will prioritize to visit the vertex having the smaller index first ($v < v'$).



Question 7: (1 point)

Choose the BEST answer below.

Suppose you run BFS and DFS starting from the same vertex v on a **complete undirected graph** with n vertices (n ≥ 2). Which of the following statements is **always true**? A complete graph is one in which every pair of distinct vertices is connected by exactly one edge.

(A) BFS and DFS will visit the vertices in exactly the same order.

(B) Both BFS and DFS trees will have depth exactly n−1.

(C) The BFS tree will have depth 1, while the DFS tree may have depth n−1.

(D) The number of edges traversed during BFS will be more than during DFS.

Student ID (SID):
Student's Name:

Question 8: (1 point)

Choose the BEST answer below:

Given an undirected, weighted graph, where V is the number of vertices and E is the number of edges in the graph. The graph does not contain any cycles. The best time complexity of the shortest path algorithm will be:

(A) O(V)

(B) $O(E^2)$

(C) $O(V^2)$

(D) O(VlogV)

(E) None of above

Question 9: (1 point)

Choose the correct statement about Breadth-First Search (BFS):

(A) BFS always explores all children of a node before moving to its grandchildren.

(B) BFS uses a stack to manage the order of vertex exploration.

(C) BFS can get stuck in an infinite loop even in a finite graph if implemented correctly.

(D) BFS is best suited for finding the shortest paths in weighted graphs.

Question 10: (1 point) **(for graduate students only)**

Choose the BEST answer below:

During a DFS traversal of a directed graph, suppose vertex u is discovered before vertex v, but v finishes before u. Which of the following statements must be true?

(A) u is an ancester of v in the DFS tree.

(B) u is a descendant of v in the DFS tree.

(C) u and v belong to different DFS trees.

(D) u and v are siblings in the DFS tree.

(E) None of above

Question 11: (1 point)

Choose the BEST answer below:

Which of the following is a true property of Breadth-First Search (BFS) when run on an unweighted connected graph starting from a source node *s*?

(A) BFS always finds the path with the minimum total edge weight from *s* to every other node.

(B) BFS guarantees finding the fewest number of edges from *s* to every reachable node.

(C) BFS might revisit the same node multiple times.

(D) The BFS traversal tree may contain cycles.

Question 12: (1 point) **(for graduate students only)**

Given a square matrix A of size $n \times n$, where each cell is labeled either 0 (reachable) or 1(blocked), we want to perform A* search for a shortest path from cell $(1,1)$ to (n,n). Rows and columns are indexed starting from 1. If you are located at a cell (x,y), you are allowed to go to neighbor cells, i.e., $(x-1,y)$, $(x+1,y)$, $(x,y-1)$, $(x,y+1)$, as long as the neighbor cell are inside the matrix. We also assume there is always a path (solution) from $(1,1)$ to (n,n). Assume that $h(x,y)$ is the heuristic that estimates the cost from (x,y) to (n,n). Which of the following heuristics is admissible?

(A) $h(x,y) = \sqrt{(n-x)^2 + (n-y)^2}$

(B) $h(x,y) = x + y$

(C) $h(x,y) = 2n - x - y$

(D) $h(x,y) = x * y$

Question 13: (1 point)

Choose the BEST answer below:

Which of the following statements correctly distinguishes between Informed and Uninformed search algorithms?

(A) Uninformed search algorithms rely on heuristic information to guide the search, while informed search algorithms do not.

(B) Informed search algorithms use additional problem-specific knowledge (heuristics) to find solutions more efficiently, while uninformed search algorithms explore the search space without such guidance.

(C) Both informed and uninformed search algorithms always guarantee finding the shortest path if one exists.

(D) Uniform Cost Search is an example of an informed search algorithm because it uses cost to prioritize nodes.

Question 14: (1 point)

Choose the BEST of these answers:

What is the following informed search that can guarantee finding the optimal solution?

(A) A* Search

(B) A Search

(C) Uniform Cost Search

(D) Greedy (Best-First) Search

(E) Both (A) and (C)

Question 15: (1 point)

In A* Search, considering the maximization problem, what is the condition of the heuristic function to be admissible?

Student ID (SID):
Student's Name:

Question 16: (1 point) **(for graduate students only)**

Given a weighted, undirected graph G (where the weight of each edge is a positive number), we have a UCS algorithm that finds the path from u to v minimizing the sum of edge weights. However, we now want to find a path that minimizes the product of edge weights instead. Assuming the Uniform Cost Search (UCS) algorithm cannot be modified, how should we transform G so that running UCS on the modified graph correctly identifies the path with the minimal product of weights?

For example, in the graph shown in Figure 1, UCS finds the shortest path from node 1 to node 6 as $1 \rightarrow 5 \rightarrow 6$ with a total cost of $2 + 1 = 3$; however, the path minimizing the product of weights is $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 6$ with a total cost of $1\times1\times1\times1=1$.

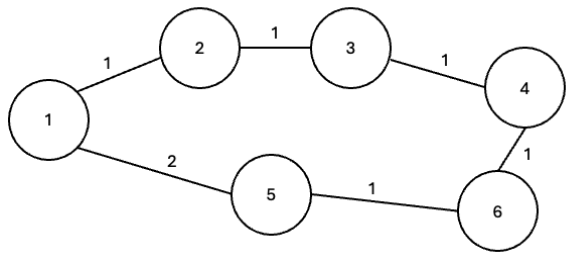


Figure 1: Example of Graph G

Question 17: (1 point)

Which of the following algorithms guarantees finding the optimal solution? Choose all correct answers.

- ☐ Uniform Cost Search
- ☐ A Search
- ☐ Greedy (Best-First) Search

☐ A* Search

Question 18: (1 point)

Choose the BEST of these answers:

In the A* search algorithm (for the minimization problem), what does the evaluation function $f(n)$ represent for a node n?

(A) The exact cost from the start node to the goal through n.

(B) The estimated cost of the cheapest path from the start to the goal node passing through n.

(C) The cost from the start node to n only.

(D) The heuristic estimate from n directly to the goal, ignoring all previous costs.

Question 19: (1 point)

Given the truth table of XOR as follows:

A	B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

Can the XOR operator between two bits be represented by a neural network with a single layer (without a hidden layer)? Why?

Student ID (SID):
Student's Name:

Question 20: (1 point)

Given a value X and a value Y where $X \in \{0, 1\}$ and $Y \in \{0, 1, 2\}$, the operator $F(X, Y)$ is defined as follows:

X	Y	$F(X,Y)$
0	0	1
0	1	1
0	2	1
1	0	1
1	1	0
1	2	0

Can $F(X, Y)$ operator be represented by the neural network with a single layer (without hidden layer)? Why?

Question 21: (1 point)

Which function(s) of the following functions are a non-linear activation? Choose all the correct answers.

- ☐ y = sin(x)
- ☐ y = e^x
- ☐ y = $\beta^n x$, where β and n are constant numbers
- ☐ y = x
- ☐ y = tanh(x)
- ☐ y = max(x, 0)

Student ID (SID):
Student's Name:

Question 22: (1 point)

Given a non-zero value x ($x \neq K$, K is a constant number), an activation function is defined as $Act(x) = \begin{cases} x \text{ if } x > K \\ 0 \text{ if } x < K \end{cases}$. The derivative of Act function can be defined as:

(A) $Act'(x) = 1$

(B) $Act'(x) = 0$

(C) $Act'(x) = \begin{cases} 0 \text{ if } x > K \\ 1 \text{ if } x < K \end{cases}$

(D) $Act'(x) = \begin{cases} 0 \text{ if } x < K \\ 1 \text{ if } x > K \end{cases}$

(E) None of above

Question 23: (1 point)

Which of the following binary operators can be represented by a single-layer neural network (without a hidden layer)? Choose all the correct answers.

☐ OR

☐ NOT

☐ AND

☐ XOR

Question 24: (1 point)

Which of the following statements about Gradient Descent is correct?

(A) Gradient Descent only needs a few iterations to converge.

(B) Gradient Descent always guarantees finding the global minimum, even for non-convex loss functions.

(C) A large learning rate can cause Gradient Descent to overshoot and possibly diverge.

(D) Gradient Descent stops immediately when the gradient at the current point is nonzero.

Question 25: (1 point)

In Gradient Descent, the learning rate controls:

(A) The number of iterations

(B) The direction of the gradient

(C) The size of steps toward the minimum

(D) None of above

Question 26: (1 point)

What is the primary goal of backpropagation?

(A) To initialize the weights

(B) To compute the gradient of model parameters with respect to the loss function

(C) The reduce the model size

(D) None of above

Question 27: (1 point)

Which of the following is true about the gradient at a local minimum?

(A) The gradient is large

(B) The gradient is negative

(C) The gradient is positive

(D) The gradient is zero

Question 28: (1 point)

In a fully connected layer, what is updated during backpropagation?

(A) Weights and biases

(B) Input data

(C) Weights only

(D) Biases only

Student ID (SID):
Student's Name:

Question 29: (1 point)

Given the L2 Loss below, y = 1, $\hat{y}$ = 3,

Choose ONE of these answers:

(A) $\ell = 1$

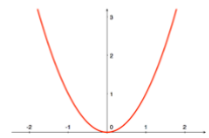
(B) $\ell = 2$

(C) $\ell = 4$

(D) None of them

L2 Loss

$\ell(\hat{y}, y) = (\hat{y} - y)^2$



Question 30: (1 point)

Given the L1 Loss below, y = 1, $\hat{y}$ = 3,

choose ONE of these answers:

(A) $\ell = 1$

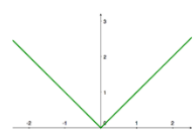
(B) $\ell = 2$

(C) $\ell = 4$

(D) None of them

L1 Loss

$\ell(\hat{y}, y) = |\hat{y} - y|$



Question 31: (1 point)

Given the Zero-One Loss below, y = 3, $\hat{y}$ = 3,

choose ONE of these answers:

(A) $\ell = 1$

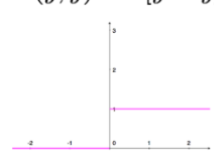
(B) $\ell = 2$

(C) $\ell = 4$

(D) None of them

Zero-One Loss

$\ell(\hat{y}, y) = \mathbf{1}[\hat{y} = y]$



Student ID (SID):
Student's Name:

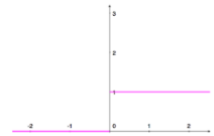
Question 32: (1 point)

Given the Zero-One Loss below, y = 1, $\hat{y}$ = 3,

choose ONE of these answers:

Zero-One Loss

$\ell(\hat{y}, y) = \mathbf{1}[\hat{y} = y]$



(A) $\ell = 1$

(B) $\ell = 2$

(C) $\ell = 4$

(D) None of them

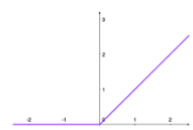
Question 33: (1 point)

Given the Hinge Loss below, y = 1, $\hat{y}$ = 2,

choose ONE of these answers:

Hinge Loss

$\ell(\hat{y}, y) = \max(0, 1 - y \cdot \hat{y})$



(A) $\ell = 1$

(B) $\ell = -1$

(C) $\ell = 0$

(D) None of them

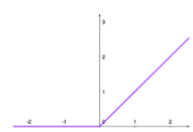
Question 34: (1 point)

Given the Hinge Loss below, y = 0, $\hat{y}$ = 1,

choose ONE of these answers:

Hinge Loss

$\ell(\hat{y}, y) = \max(0, 1 - y \cdot \hat{y})$



(A) $\ell = 1$

(B) $\ell = -1$

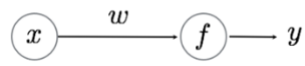
(C) $\ell = 0$

(D) None of them

Student ID (SID):
Student's Name:

Question 35: (1 point)

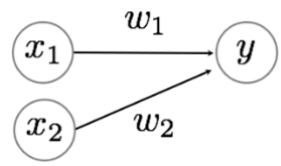
Given the World's Smallest Perceptron, i.e., $\hat{y} = wx$, and its derivative.



Let us define $L = \frac{1}{2}(y - \hat{y})^2$. Please derive the gradient $\nabla w = \frac{\partial L}{\partial w}$.

Question 36: (1 point) **(for graduate students only)**

Given the World's Second Smallest Perceptron, i.e., $\hat{y} = w_1 x_1 + w_2 x_2$. Let us define $L = \frac{1}{2}(y - \hat{y})^2$. Please derive the gradient $\nabla w_2 = \frac{\partial L}{\partial w_2}$



Question 37: (1 point)

Choose the BEST answer:

Why should we avoid initializing all weights and biases to zero in a neural network?

(A) It would make the training much faster but less accurate.

(B) All neurons would learn the same features during training, preventing the network from breaking symmetry.

(C) It would randomly freeze half of the neurons during forward propagation.

(D) It would cause the loss function to become non-differentiable.

Question 38: (1 point)

Choose the BEST answer:

How do neural networks learn the values of their weights and biases during training?

(A) By randomly reinitializing weights and biases after every training example.

(B) By manually setting the values based on human intuition.

(C) By utilizing an optimization algorithm such as Gradient Descent to minimize a loss function through iterative updates.

(D) By selecting the highest values of weights and biases from a predefined list at the end of each epoch.

Question 39: (1 point)

Why does the activation function need to be non-linear?

Question 40: (1 point)

Choose the BEST answer:

When we add more layers to a neural network without increasing the amount of training data, which of the following issues is most likely to occur?

(A) The model becomes simpler and underfits the data.

(B) The model may overfit the training data and generalize poorly to new data.

(C) The model trains faster because deeper networks converge more quickly.

(D) The model automatically requires fewer parameters to learn complex patterns.

Question 41: (1 point)

How is the Backward Propagation different from Forward Propagation?

Question 42: (1 point)

Choose the BEST answer.

Which of the following is a direct consequence of poor weight initialization in a deep neural network?

(A) Faster convergence to the optimal solution.

(B) Slower or failed convergence due to vanishing/exploding gradients.

(C) Guaranteed better generalization to unseen data.

(D) Immediate overfitting of the training dataset.

Student ID (SID):
Student's Name:

Question 43: (1 point)

In the context of neural networks, what are derivatives used for?

(A) Updating the weights during training

(B) Initializing the weights

(C) Reducing dimensionality

(D) None of above

Question 44: (1 point)

What is the derivative of the ReLU function $f(x) = max(0, x)$?

(A) Always 1

(B) 0 for $x > 0$, 1 for $x \leq 0$

(C) 1 for $x > 0$, 0 for $x \leq 0$

(D) x

Question 45: (1 point)

What is the correct formula to calculate Bayesian probability?

(A) P(B|A) = (P(A|B) * P(B)) / P(A)

(B) P(A|B) = (P(B|A) + P(A)) / P(B)

(C) P(A|B) = (P(A) * P(B)) / P(B|A)

(D) P(B|A) = (P(A) + P(B)) / P(A|B)

(E) None of above

Question 46: (1 point) **(for graduate students only)**

You toss a fair coin three times. What is the probability that you get exactly two heads given that the first toss is a head?

(A) 1/4

(B) 1/3

(C) 1/2

(D) 2/3

(E) None of above

Question 47: (1 point)

Given a learning rate α, parameter w, a learning objective L, what is the correct formula to update the parameter w in the gradient descent algorithm? Choose the BEST answer.

(A) $w = w + \alpha \frac{\partial L}{\partial w}$

(B) $w = w - \alpha \frac{\partial L}{\partial w}$

(C) $w = w + \frac{\partial L}{\partial w}$

(D) $w = w - \frac{\partial L}{\partial w}$

(E) None of above

Question 48: (1 point)

Choose the BEST answer.

Which of the following statements about neural networks is correct?

(A) A single-layer neural network without an activation function can model any nonlinear relationship.

(B) Increasing the number of layers and neurons always guarantees better performance.

(C) Neural networks learn by adjusting weights and biases to minimize a loss function using optimization algorithms like Gradient Descent.

(D) Neural networks never suffer from overfitting, even with small datasets.

Question 49: (1 point)

Choose the BEST answer.

Which of the following statements about activation functions in neural networks is correct?

(A) Activation functions introduce non-linearity, allowing neural networks to model complex relationships.

(B) Without activation functions, a neural network would behave like a simple linear model, regardless of its depth.

(C) The ReLU activation function outputs zero for all negative input values. ReLU(x) = max(x, 0).

(D) All of the above.

Question 50: (1 point)

Which of the following is an example of a Constraint Satisfaction Problem (CSP)? Choose all the correct answers.

- ☐ Classifying images into different categories using a neural network.
- ☐ Scheduling final exams for university courses without overlapping student exams.
- ☐ Solving Sudoku puzzles by assigning numbers to empty cells without violating the rules.
- ☐ Finding the minimum of a non-convex function using gradient descent.